

JEE MAIN REVISION MASTER: CHEMICAL EQUILIBRIUM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 118 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Equilibrium Constants (K_p , K_c) & Reaction Quotient	96 PYQs	Equilibrium Constants
Very High	Le Chatelier's Principle & Stress Factors	22 PYQs	Le Chatelier

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Equilibrium Constants (K_p , K_c) & Reaction Quotient	$K_p = K_c(RT)^{\Delta n_g}$, $\Delta G^\circ = -RT \ln K = -2.303RT \log_{10} K$, $Q < K \implies \text{Forward}$
Le Chatelier's Principle & Stress Factors	$\ln \left(\frac{K_2}{K_1} \right) = \frac{\Delta H^\circ}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$ (van 't Hoff)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Equilibrium Constants (K_p , K_c) & Reaction Quotient (Very High Yield — 96 PYQs)

Core Principle: Relationship between K_p and K_c , expression of equilibrium constants, and spontaneous direction based on Q vs K .

■■ **Caution / JEE Pitfall:** In calculating Δn_g , count ONLY gaseous species: $\Delta n_g = \sum n_{\text{(gas)}}(\text{products}) - \sum n_{\text{(gas)}}(\text{reactants})$. Pure solids and liquids are excluded ($a = 1$).

■ **Speed Trick:** If a reaction is multiplied by factor n , its new equilibrium constant becomes $K' = K^n$. If reversed, $K' = 1/K$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Consider the following reactions in which all the reactants and products are present in gaseous state $\begin{matrix} \text{N}_2 + 3\text{H}_2 & \rightleftharpoons & 2\text{NH}_3 \end{matrix}$..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Observe the following equilibrium in a 1 L flask. $\text{A(g)} \rightleftharpoons \text{B(g)}$ At $T(K)$, the equilibrium concentrations of A and B are 0.5 M and 0.375 M respec..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (Numerical): "For the following gas phase equilibrium reaction at constant temperature, $\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g})$ if the tota..." [Ans: 125]

3.2 Le Chatelier's Principle & Stress Factors (Very High Yield — 22 PYQs)

Core Principle: Dynamic response of equilibrium systems to variations in concentration, temperature, pressure, and inert gas addition.

■ **Caution / JEE Pitfall:** Addition of an inert gas at CONSTANT VOLUME has NO effect on equilibrium position because partial pressures remain unchanged!

■ **Speed Trick:** Inert gas addition at CONSTANT PRESSURE shifts equilibrium toward the side with MORE gaseous moles ($\Delta n_g > 0$ shifts forward).

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (Numerical):** "In a closed flask at 600 K, one mole of $X_2Y_4(g)$ attains equilibrium as given below: $X_2Y_4(g) \rightleftharpoons 2XY_2(g)$..." [Ans: 8]
- **[2026 Evening Shift] Q#2 (MCQ):** "Consider the following gaseous equilibrium in a closed container of volume 'V' at T(K). $P_2(g) + Q_2(g) \rightleftharpoons 2PQ(g)$..." [Ans: N/A]
- **[2025 Evening Shift] Q#3 (MCQ):** "Consider the equilibrium $CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$ If the pressure applied over the system increases by two f..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

In a closed flask at 600 K, one mole of $X_2Y_4(g)$ attains equilibrium as given below: $X_2Y_4(g) \rightleftharpoons 2XY_2(g)$ At equilibrium, 75% $X_2Y_4(g)$ was dissociated and the total pressure is 1 atm. The magnitude of $\Delta_r G^\circ$ (in kJ mol^{-1}) at this temperature is _____. (Nearest Integer) (Given: $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$; $\ln 10 = 2.3$, $\log 2 = 0.3$, $\log 3 = 0.48$, $\log 5 = 0.69$, $\log 7 = 0.84$)

Correct Answer: 8

Solution Analysis: $P^\circ(1+\alpha) = 1$ $\alpha = (3/4)$; $P^\circ = (4/7)$ $K_p = \frac{4P_o}{\alpha^2} = \frac{36}{7}$ $\Delta G^\circ = -RT \ln K_p$ $|\Delta G^\circ| = 8.314 \times 600 \ln((36/7)) = 8169.5 \text{ J mol}^{-1} = 8.169 \text{ kJ mol}^{-1}$

Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Morning Shift (Q#1)

For the following gas phase equilibrium reaction at constant temperature, $NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$ if the total pressure is $\sqrt{3}$ atm and the pressure equilibrium constant (K_p) is 9 atm, then the degree of dissociation is given as $(x \times 10^{-2})^{1/2}$. The value of x is _____. (nearest integer)

Correct Answer: 125

Solution Analysis: $K_p = \frac{(\alpha/2)^{1/2}((3\alpha/2)^{3/2}(1-\alpha))}{P(T)(1+\alpha)^{1/2}}$ $P(T) = \sqrt{3} \text{ atm}$ $\Rightarrow 9 = \frac{(\alpha/2)^{1/2}((3\alpha/2)^{3/2}(1-\alpha))}{\sqrt{3}(1+\alpha)^{1/2}}$ $\Rightarrow 9 = \frac{9}{\sqrt{3}} \frac{\alpha^{1/2}(1-\alpha)^{3/2}}{(1+\alpha)^{1/2}}$ $\Rightarrow 1-\alpha^2 = \frac{5\alpha^2}{1+\alpha^2}$ $\Rightarrow \alpha^2 = 0.8$ $\Rightarrow \alpha = (0.8)^{1/2} \Rightarrow \alpha = [(1/0.8)]^{1/2}$ $\alpha = [125 \times 10^{-2}]^{1/2}$ $\therefore x = 125$.